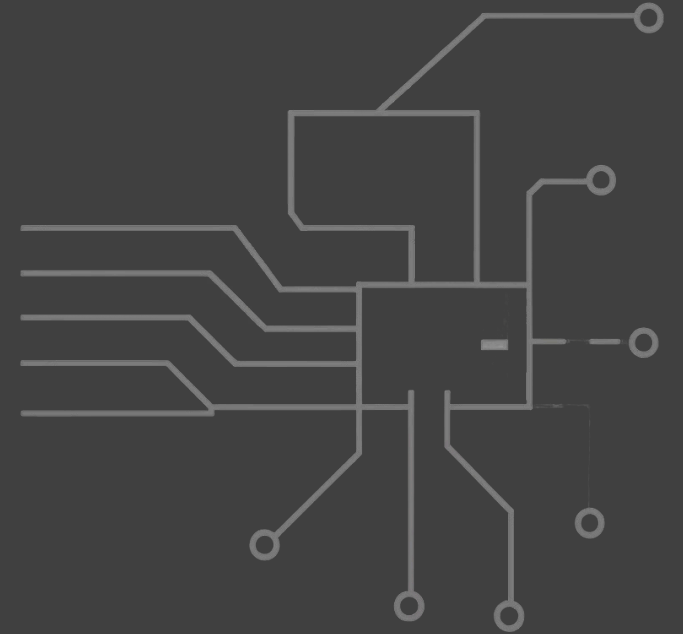


Asynchronous Dataflow

The Evolution of High-Performance Systems



Beyond Control-Driven Models

Traditional Execution

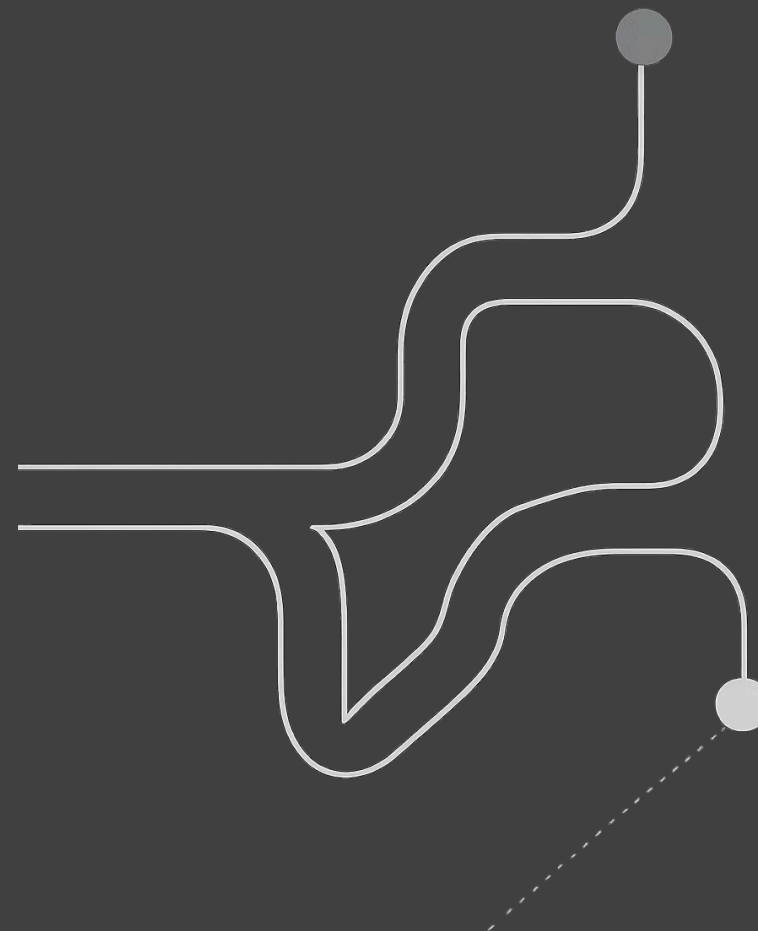
Thread blocking & sequential bottlenecks.

The Dataflow Paradigm

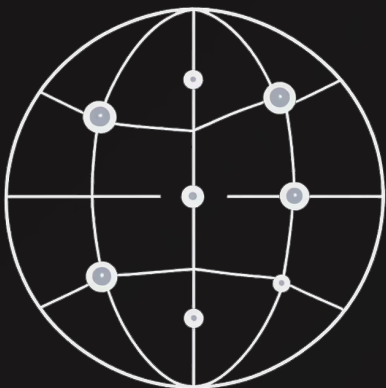
Execution triggers strictly upon data availability.

Impact

Unlocks massive parallelism without arbitrary ordering.



The Architectural Shift



Legacy Bottleneck

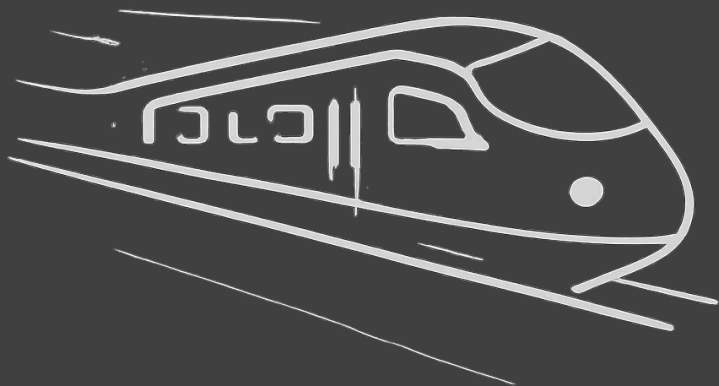
Synchronous barriers & batch processing overhead (e.g., legacy Spark/MapReduce).

The Breakthrough

Decoupling **control-plane latency** from data-plane execution

Core Innovation

Asynchronous operator dispatch driven by data arrival.



Naiad & the Timely Dataflow Revolution

1

Hybrid Execution

Unifying fine-grained **sync + async** processing.

2

Progress Tracking

Structured timestamps manage **cyclic dataflow** — no deadlock

3

Performance Leap

Sub-millisecond iterations bypassing legacy batch processing

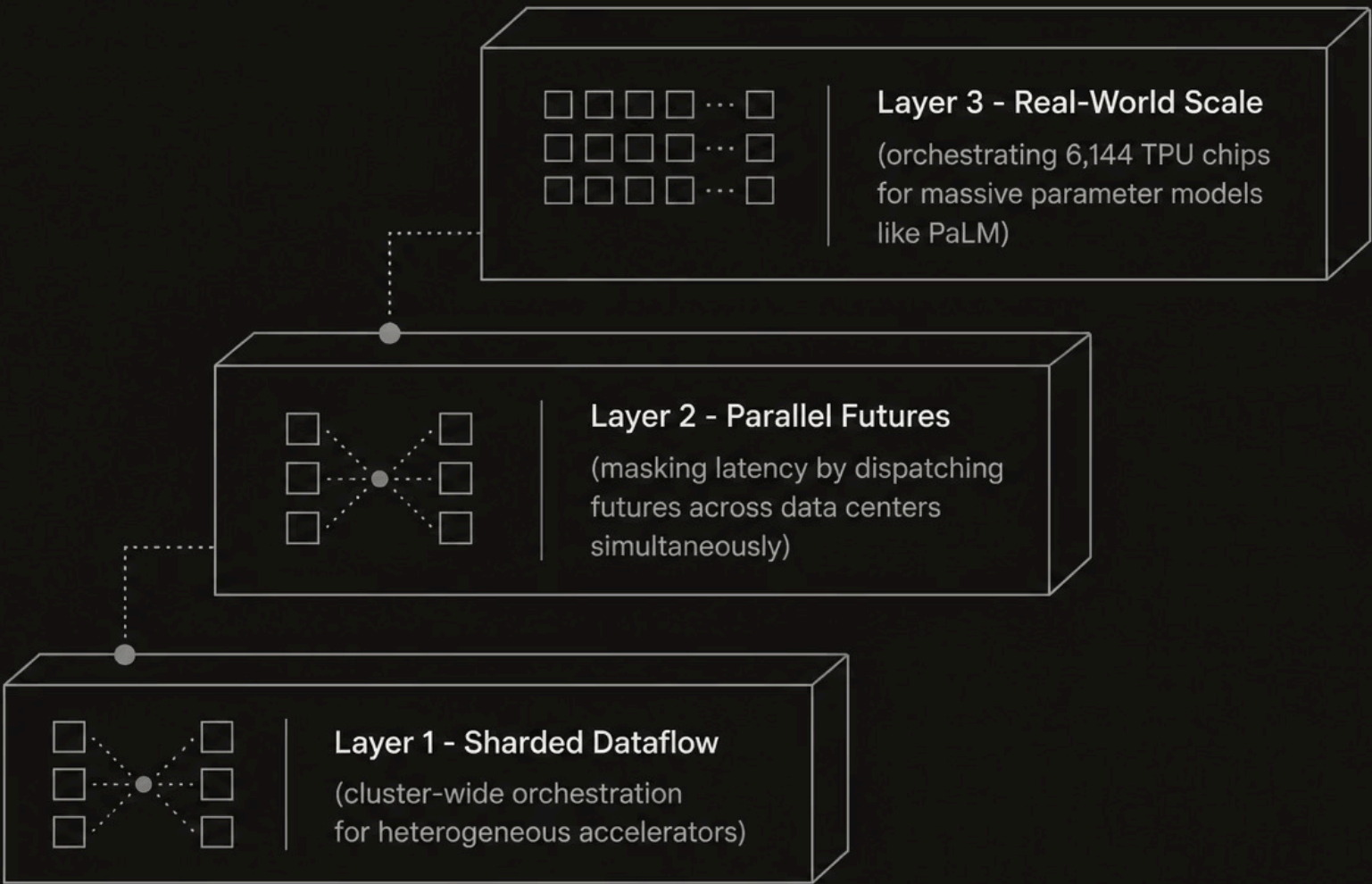
Modern Orchestration: Pathways

6,144

TPU Chips

Orchestrated for PaLM-scale models

Sharded dataflow model enabling **cluster-wide orchestration** across heterogeneous accelerators.



Application Case Study

Real-Time IoT Sensor Data Processing

1

The Software Bridge

Bringing the "Dataflow Rule" to the application layer, ensuring execution triggers only upon data availability.

2

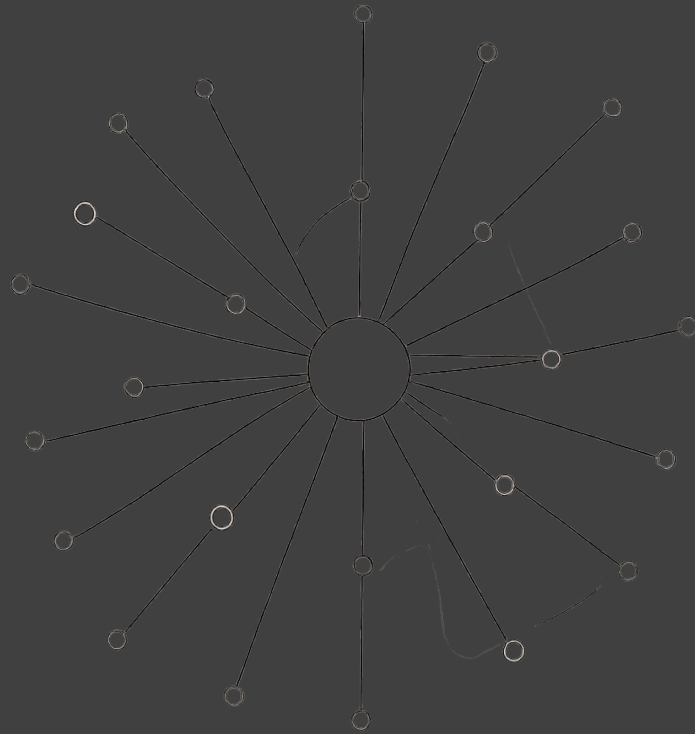
Tech Stack

TypeScript, **Node.js**, and **Serverless PostgreSQL** (Neon/Supabase) provide a modern, scalable foundation.

3

Core Mechanics

Leveraging **Async Generators** (`async function*`) & **Async Iterators** (`for await...of`) for efficient, non-blocking data consumption.



The Future of Data-Driven Systems

Convergence

Timeliness + Consistency +
Expressivity

Next Era

Virtualized async dataflow
powering **global-scale AI**

Core Takeaway

Align architecture with **data's natural flow** — unlock unprecedented scale